OPERATING SYSTEMS + LAB

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COURSE CODE: CS2701

COURSE OBJECTIVES

The objective of this course is to enable the student to:

- To study the basic concepts and functions of operating systems.
- To understand the structure and functions of OS.
- To learn about processes, threads and scheduling algorithms.
- To understand the principles of concurrency and Deadlocks.
- To learn various memory management schemes, I/O management and file system implementations.

INTRODUCTION TO OPERATING SYSTEMS **7**

Introduction: Computer system organization - Computer system architecture - Operating system operations, Operating system structures: Operating system Services - System calls - System programs - Operating system structure - Operating system generation - System Boot.

PROCESSES AND THREADS **10**

Processes: Process concept - Process scheduling - Operations on processes - Interprocess communication, Threads: Multi core programming - Multithreading models - Threading issues, CPU Scheduling: Basic concepts - Scheduling criteria - Scheduling algorithms - Thread scheduling - Multiple processor scheduling - Real time CPU scheduling.

CONCURRENCY **9**

Process Synchronization: Background - The Critical Section problem - Peterson's solution - Synchronization hardware - Mutex Locks - Semaphores - Classic problems of synchronization – Monitors, Deadlocks: System model - Deadlock characterization - Methods for handling deadlocks: Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from deadlock.

MEMORY MANAGEMENT **9**

Main Memory: Background – Swapping - Contiguous Memory Allocation – Segmentation – Paging - Structure of the Page Table – Virtual Memory: Background - Demand Paging - Copy-on-Write - Page Replacement - Allocation of Frames – Thrashing - Memory-Mapped Files - Allocating Kernel Memory.

STORAGE MANAGEMENT **10**

Mass-Storage Structure: Disk Structure - Disk Scheduling - Disk Management - Swap-Space Management - I/O Systems: I/O Hardware - Application I/O Interface - Kernel I/O Subsystem - File-System Interface: File concept - Access methods - Directory and Disk Structure - File-System Implementation: File-System Structure - File-System implementation
- Directory implementation - Allocation methods - Free-Space management.

TOTAL HOURS : 45

COURSE OUTCOMES

On successful completion of this course, the student will be able to:

- Understand the importance of operating system.
- Apply the principles of concurrency.
- Appreciate process synchronization, deadlock avoidance and deadlock prevention mechanisms.
- Compare and contrast various memory management schemes.
- Understand the various file systems and file management mechanisms.

TEXTBOOKS

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, Operating System Concepts, 9th Edition, John Wiley and Sons Inc., 2012.

<u>LABORATORY EXERCISES</u>

SUGGESTIVE LIST OF EXPERIMENTS:

1. Linux OS level scripting.
2. System admin and task automation in Linux.
3. Implementation of Inter Process Communication (IPC) using Shared Memory.
4. Implementation of Inter Process Communication (IPC) using Pipes.
5. Implementation of CPU Scheduling Algorithms.
6. Creation of Threads and Synchronization Applications.
7. Implementation of Process Synchronization, Deadlock Avoidance and Detection Mechanisms.